# LATENT SPACE OF VAE

**Viacheslav Chaunin, Alina Ermilova, Anastasiia Chernysheva***

**Project Supervisor: Alexander Kolesov**

Skolkovo Institute of Science and Technology, Skoltech

## 1 INTRODUCTION

Generative modeling represents a fundamental component of modern machine learning, enabling systems to learn data distributions and generate novel samples. **V**ariational **A**uto**e**ncoders (VAEs) (Kingma & Welling, 2014) serve as probabilistic generative models that learn interpretable latent representations by mapping complex data distributions to simpler latent spaces. Understanding the geometric properties of these latent spaces is crucial for improving generation quality and model robustness. This project investigates how different VAE variants, namely, the classical VAE, **I**mportance **W**eighted **A**uto**e**ncoders (IWAE) (Burda et al., 2016), and VAE with **Va**riational **M**ixture of **P**osteriors **Prior** (VampPrior VAE) (Tomczak & Welling, 2018), structure their latent spaces and examines the impact of these geometric properties on sampling quality.

## 2 RELATED WORK

The standard VAE architecture, introduced by Kingma & Welling (2014), learns compressed latent representations but suffers from critical structural defects including posterior collapse, holes in the latent space, and inadequate posterior coverage of the prior distribution (Burda et al., 2016). The authors of Burda et al. (2016) address these issues by introducing the IWAE model, which maximizes a tighter lower bound on the marginal likelihood through multi-sample evidence lower bounds. However, this approach can be computationally expensive and prone to overfitting (Tomczak & Welling, 2018). The VampPrior VAE by Tomczak & Welling (2018) introduces a mixture of variational posteriors with pseudo-inputs, decomposing the training objective into reconstruction error, entropy of the variational posterior, and cross-entropy between the aggregated posterior and the prior.

While all these modifications exist and are widely used, their impact on latent space geometry remains underexplored in the literature. We expect to fill this gap using Colored MNIST classification task.

## 3 METHODOLOGY

### 3.1 DATASET

For the experiments, we use the Colored MNIST dataset[1], which extends the classical MNIST dataset by assigning unique colors to each digit. We utilize training batch size equals to $64$, while the validation one is $256$, with input images resized to $32 \times 32$ pixels. This dataset provides a controlled environment for evaluating generative models, while introducing additional complexity through color variation, making it suitable for analyzing latent space structure and generation quality across different model architectures.

Figure 1 shows several examples from our dataset.

*All project team members contributed equally

[1] Provided by Alexander Kolesov, `https://github.com/ngushchin/EntropicNeuralOptimalTransport/blob/main/src/tools.py`

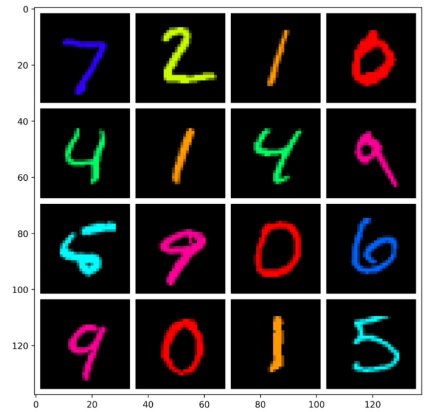


Figure 1: Sample from the Colored MNIST dataset.

### 3.2 METHODS

In our experiments, we examine the three models: VAE, IWAE, and VampPrior VAE.

#### 3.2.1 VAE

The Variational Autoencoder (Kingma & Welling, 2014) learns stochastic mappings between an observed data space $\mathbf{x}$ with complex empirical distribution $q_D(\mathbf{x})$ and a latent space $\mathbf{z}$ with relatively simple distribution structure, typically Gaussian. The generative model learns a joint distribution $p_\theta(\mathbf{x}, \mathbf{z})$ that is factorized as $p_\theta(\mathbf{x}, \mathbf{z}) = p_\theta(\mathbf{z})p_\theta(\mathbf{x}|\mathbf{z})$, where $p_\theta(\mathbf{z})$ represents the prior distribution over latent space and $p_\theta(\mathbf{x}|\mathbf{z})$ is the stochastic decoder. The encoder network, also called the inference model $q_\phi(\mathbf{z}|\mathbf{x})$, approximates the true but intractable posterior $p_\theta(\mathbf{z}|\mathbf{x})$ of the generative model. Figure 2 shows the architecture of the VAE model and Figure 3 represents its computational flow.

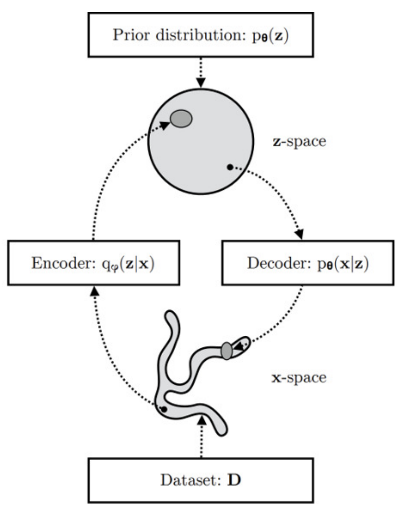



Figure 2: VAE architecture showing the mapping between observed $\mathbf{x}$-space and latent $\mathbf{z}$-space through encoder and decoder networks (Kingma et al., 2019).

The VAE is trained by maximizing the Evidence Lower Bound (ELBO), which provides a tractable objective for optimization:

$$\text{ELBO} = \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}\left[\log p_\theta(\mathbf{x}|\mathbf{z})\right] - D_{KL}\left(q_\phi(\mathbf{z}|\mathbf{x}) \| p_\theta(\mathbf{z})\right) \tag{1}$$

The first term represents the reconstruction likelihood, encouraging the decoder to accurately reconstruct input data from latent codes sampled from the encoder distribution. The second term is the

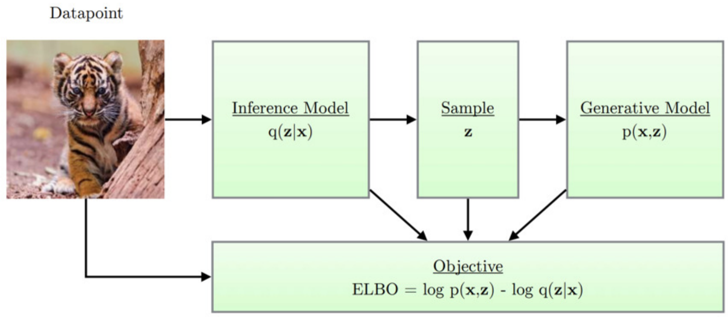



Figure 3: Computational flow in a variational autoencoder showing the relationship between encoder (inference model), decoder (generative model), and the ELBO objective (Kingma et al., 2019).

KL divergence between the approximate posterior and the prior, acting as a regularization that encourages the learned posterior to match the prior distribution. This objective balances reconstruction accuracy with the constraint that latent representations should follow a structured prior distribution, typically a standard Gaussian $\mathcal{N}(0, I)$.

However, VAEs suffer from several pathologies including posterior collapse, where latent dimensions become independent of the data and revert to the prior, holes in the latent space that decode to unrealistic samples, and poor coverage where the learned posterior does not adequately span the prior distribution. These issues arise from the balance between reconstruction and regularization in the ELBO objective, often resulting in underutilized latent capacity and suboptimal generation quality.

#### 3.2.2 IWAE

The Importance Weighted Autoencoder (Burda et al., 2016; Cremer et al., 2017) addresses the issues of the standard VAE's evidence lower bound by utilizing importance sampling with multiple latent samples. Instead of using a single sample from the encoder distribution, IWAE draws $K$ samples $\{\mathbf{z}_i\}_{i=1}^{K}$ and computes an importance-weighted estimate of the log marginal likelihood. This approach provides a tighter lower bound than the standard ELBO, as the multi-sample evidence lower bound converges to the true marginal likelihood as $K \to \infty$.

The IWAE maximizes the following multi-sample evidence lower bound:

$$\log p(\mathbf{x}) \geq \mathbb{E}_{\mathbf{z}_{1:K} \sim q(\mathbf{z}|\mathbf{x})} \left[ \log \left( \frac{1}{K} \sum_{i=1}^{K} \frac{p(\mathbf{x}, \mathbf{z}_i)}{q(\mathbf{z}_i|\mathbf{x})} \right) \right] = L_{\text{IWAE}}[q] \tag{2}$$

This bound is provably tighter than the standard VAE ELBO, meaning $L_{\text{IWAE}}[q] \geq L_{\text{VAE}}[q]$ for all $K \geq 1$. Alternatively, IWAE can be viewed as optimizing the standard variational lower bound but using a more complex, expressive distribution family for the approximate posterior. By incorporating multiple samples, IWAE pushes the approximate posterior $q(\mathbf{z}|\mathbf{x})$ closer to the true posterior $p(\mathbf{z}|\mathbf{x})$, potentially resolving some of the coverage issues present in standard VAEs.

While IWAE achieves better likelihood bounds and can produce higher-quality density estimates, it comes with significant computational overhead proportional to the number of importance samples $K$. The model is also susceptible to overfitting, as the tighter bound can lead to over-peaked posteriors that concentrate probability mass too narrowly around training data.

#### 3.2.3 VampPrior VAE

The VampPrior VAE (Tomczak & Welling, 2018) restructures the variational lower bound to enhance model flexibility and address limitations of standard priors. The training objective is decomposed into three distinct components:

$$\mathcal{L}(\phi, \theta, \lambda) = \underbrace{\mathbb{E}_{\mathbf{x} \sim q(\mathbf{x})} \left[ \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})} \left[ \ln p_\theta(\mathbf{x}|\mathbf{z}) \right] \right]}_{\text{negative reconstruction error}} + \underbrace{\mathbb{E}_{\mathbf{x} \sim q(\mathbf{x})} \left[ H[q_\phi(\mathbf{z}|\mathbf{x})] \right]}_{\substack{\text{expectation of} \\ \text{posterior entropy}}} - \underbrace{\mathbb{E}_{\mathbf{z} \sim q(\mathbf{z})} \left[ - \ln p_\lambda(\mathbf{z}) \right]}_{\substack{\text{cross-entropy between} \\ \text{aggregated posterior } q(\mathbf{z}) \\ \text{and prior } p_\lambda(\mathbf{z})}} \quad (3)$$

where $q(\mathbf{z}) = \frac{1}{N} \sum_{n=1}^{N} q_\phi(\mathbf{z}|\mathbf{x}_n)$ denotes the aggregated posterior. The second term encourages high entropy (e.g., large variance) in the variational posterior for individual data points, while the third term aligns the aggregated posterior with the prior $p_\lambda(\mathbf{z})$.

To overcome issues like overfitting and computational inefficiency in standard priors, VampPrior introduces a learnable prior defined as a mixture of variational posteriors conditioned on pseudo-inputs:

$$p_\lambda(\mathbf{z}) = \frac{1}{K} \sum_{k=1}^{K} q_\phi(\mathbf{z}|\mathbf{u}_k), \quad (4)$$

where $K$ is the number of pseudo-inputs, and each $\mathbf{u}_k \in \mathbb{R}^D$ is a learnable $D$-dimensional vector. This construction allows the prior to adapt to the data distribution through the pseudo-inputs $\{\mathbf{u}_1, \ldots, \mathbf{u}_K\}$, improving expressiveness while maintaining computational efficiency.

**Technical details.** We train all models using the AdamW optimizer with a learning rate of $0.001$ and weight decay of $0.0001$ for $25$ epochs.

#### 3.2.4 Langevin Dynamics

To enhance latent space exploration, we apply Langevin dynamics (Welling & Teh, 2011) after models' training.

The core algorithm can be understood as an extension of stochastic gradient methods with the addition of controlled noise:

$$\Delta\theta_t = \frac{\epsilon_t}{2} \left( \nabla \log p(\theta_t) + \frac{N}{n} \sum_{i=1}^{n} \nabla \log p(x_{ti}|\theta_t) \right) \quad (5)$$

When incorporating Langevin dynamics, we add Gaussian noise to the update and this leads to the following Stochastic Gradient Langevin Dynamics formulation:

$$\Delta\theta_t = \frac{\epsilon_t}{2} \left( \nabla \log p(\theta_t) + \frac{N}{n} \sum_{i=1}^{n} \nabla \log p(x_{ti}|\theta_t) \right) + \eta_t, \quad \eta_t \sim \mathcal{N}(0, \epsilon_t) \quad (6)$$

In our implementation, we adapt these formulas to the latent space context where $\theta$ represents latent variables $\mathbf{z}$, $p(\theta)$ corresponds to the prior $p(\mathbf{z})$, and $p(\mathbf{x}|\theta)$ represents the decoder likelihood $p_\theta(\mathbf{x}|\mathbf{z})$. The posterior distribution of interest is $p(\mathbf{z}|\mathbf{x}) \propto p(\mathbf{z})p_\theta(\mathbf{x}|\mathbf{z})$.

In our experiments, we explore how the addition of Langevin dynamics influence the generation quality. So, we compare the results of classical sequential implementation of encoder and decoder to the usage of Langevin dynamics between them: implementing Langevin dynamics on the encoder's output before applying the decoder.

### 3.3 Evaluation Quality

#### 3.3.1 Latent Representations Quality

To analyze the learned latent representations, we extract latent codes for the validation set by computing the approximate posterior parameters $\mu(x)$ and $\log \sigma^2(x)$, as well as sampled latent variables

$$z \sim q_\phi(z \mid x) = \mathcal{N}\big(z;\, \mu(x), \mathrm{diag}(\sigma^2(x))\big)\,, \quad (7)$$

using the reparameterization trick. This enables a unified comparison of latent geometry across all models.

Statistical properties of the latent space are examined using multiple complementary diagnostics. Low-dimensional structure is visualized via PCA applied to latent samples $\{z_i\}_{i=1}^N$. The effective dimensionality is quantified through the number of *active units*, defined as latent dimensions whose empirical variance exceeds a fixed threshold:

$$\mathrm{Var}_x\left[\mu_j(x)\right] > \tau. \tag{8}$$

Additionally, per-dimension Kullback–Leibler divergence is computed to assess how strongly each latent dimension deviates from the prior:

$$D_{\mathrm{KL}}(q(z_j \mid x) \,\|\, p(z_j)) \,. \tag{9}$$

To evaluate semantic separability, a *linear probe* in the form of multinomial logistic regression is trained on latent representations ($\mu(x)$ or $z$) to predict digit labels:

$$p(y \mid z) = \mathrm{softmax}(Wz + b). \tag{10}$$

High probe accuracy indicates that class-relevant information is linearly encoded in the latent space.

Local semantic consistency is further examined through latent traversals, where individual latent dimensions are perturbed while others are held fixed,

$$z_j \leftarrow z_j + \delta, \quad \delta \in \mathbb{R}, \tag{11}$$

and the resulting decoded images $g_\theta(z)$ are inspected for smooth and interpretable changes. Finally, to probe posterior geometry beyond variational approximations, Markov Chain Monte Carlo (MCMC) methods are applied in latent space after training. In particular, we use the Metropolis-adjusted Langevin algorithm (MALA) (Roberts & Tweedie, 1996) to locally refine latent points with respect to the true posterior. Starting from linear interpolations between latent codes of two data points,

$$z_t^{\mathrm{lin}} = (1-t)\, z_a + t\, z_b, \quad t \in [0,1], \tag{12}$$

short MALA chains are run to relax each interpolated point toward higher posterior density regions. Each MALA update proposes

$$z' = z + \frac{\epsilon}{2} \nabla_z \log p(z \mid x) + \sqrt{\epsilon}\, \eta, \quad \eta \sim \mathcal{N}(0, I), \tag{13}$$

and is accepted with probability

$$\alpha(z, z') = \min\left(1, \frac{p(z' \mid x)\, q(z \mid z')}{p(z \mid x)\, q(z' \mid z)}\right). \tag{14}$$

We report the MALA acceptance rate, defined as the fraction of accepted proposals along each relaxed interpolation segment, which serves as a diagnostic of posterior smoothness, connectivity, and multimodality.

### 3.3.2 Generation Quality Metrics

We evaluate the generation quality using **F**réchet **I**nception **D**istance (FID) and **K**ernel **I**nception **D**istance (KID). Both metrics utilize `Inception-v3`'s features extracted from real versus generated images and the lower they are the better is the generated images' quality.

FID measures the distance between feature distributions under Gaussian assumptions, though it is sensitive to sample size and can be biased.

KID estimates Maximum Mean Discrepancy using a polynomial kernel, providing an *unbiased* estimator that works well with small samples and detects mode collapse more effectively than FID by capturing higher-order moments.

Table 1: Reconstruction quality metrics (FID↓, KID↓) with different numbers of Langevin dynamics steps. Using 0 steps represents the case without the usage of Langevin dynamics. The best values for each model are highlighted in bold.

| Steps | VAE | | IWAE | | VampPrior VAE | |
|---|---|---|---|---|---|---|
| | FID | KID | FID | KID | FID | KID |
| 0 | 19.38 | 0.00907 | 18.78 | 0.00870 | 17.98 | 0.00814 |
| 5 | 19.21 | 0.00904 | **18.40** | **0.00869** | 18.23 | 0.00822 |
| 10 | 19.02 | 0.00905 | 18.43 | 0.00892 | 18.27 | 0.00828 |
| 20 | **18.53** | **0.00890** | 19.06 | 0.00936 | **18.04** | **0.00810** |
| 50 | 20.43 | 0.01010 | 20.97 | 0.01052 | 18.55 | 0.00873 |
| 100 | 23.67 | 0.01232 | 24.75 | 0.01299 | 18.83 | 0.00897 |

# 4 Results and Analysis

## 4.1 Generation Quality: Langevin Dynamics' Role

Table 1 presents the quantitative impact of applying Langevin dynamics on reconstruction quality across all three VAE variants. We calculate FID and KID for different numbers of Langevin steps. The lower values of FID and KID indicate better generation quality.

Our results demonstrate that Langevin dynamics significantly improves reconstruction quality when properly calibrated to each model's latent space geometry. Key findings include:

- **The best number of steps vary by architecture**: VAE and VampPrior VAE achieve best performance at $20$ steps, while IWAE – at $5$ steps, reflecting different posterior geometries that require distinct exploration strategies.
- **Non-monotonic improvement pattern**: All models initially benefit from Langevin refinement ($\approx 0 - 20$ steps), but excessive steps ($\approx 50+$) degrade quality, particularly for VAE and IWAE. This suggests that over-optimization can disrupt the natural latent space structure.
- **VampPrior's robustness**: The VampPrior VAE maintains superior performance across all step counts and shows minimal degradation even at higher step counts, confirming its more coherent latent space structure.
- **Model comparison**: At optimal step counts, VampPrior (FID $= 18.04$ and KID $= 0.00810$) outperforms both standard VAE and IWAE.

These results validate our hypothesis that different VAE variants create latent spaces with distinct geometric properties. The superior performance of VampPrior VAE with Langevin dynamics confirms that its pseudo-input-based prior creates a more structured and explorable latent space, making it particularly suitable for applications requiring high-quality generation and reliable interpolation.

## 4.2 Latent Space Structure and Posterior Geometry

We now compare the latent representations learned by VAE, IWAE, and VampPrior VAE using the diagnostics introduced in the previous section. Our analysis focuses on global structure, semantic separability, and local posterior geometry.

**Statistical structure and utilization.** All three models exhibit a high number of active latent dimensions. VAE and VampPrior utilize all $16/16$ dimensions, while IWAE uses $15/16$, indicating no severe posterior collapse. However, a more fine-grained view provided by the per-dimension KL divergence (Figure 4) reveals qualitative differences in how information is distributed across the latent space. VAE and IWAE allocate KL relatively uniformly across dimensions, suggesting more homogeneous latent usage. In contrast, VampPrior exhibits several dimensions with markedly higher KL values, indicating stronger specialization and uneven information concentration induced by the learned mixture prior.

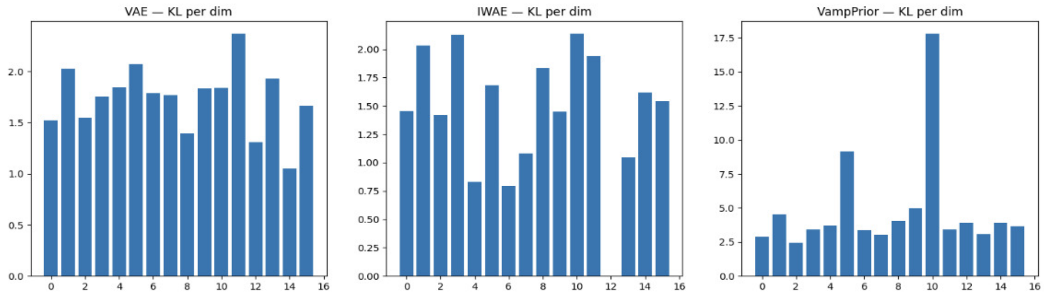



Figure 4: Per-dimension KL divergence for VAE, IWAE, and VampPrior.

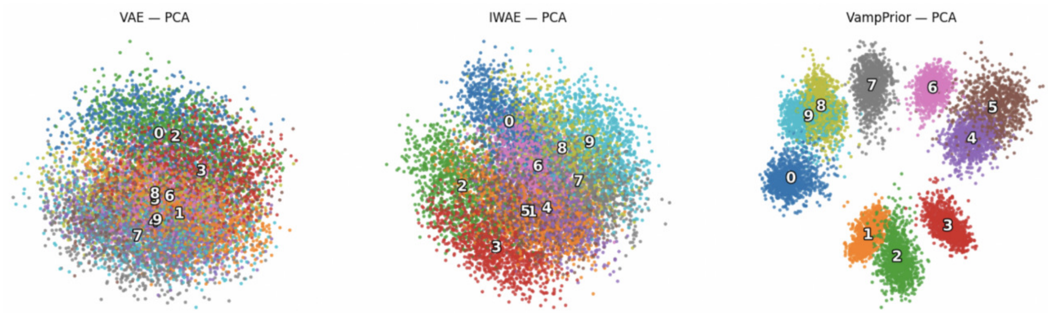



Figure 5: PCA projections of latent representations for VAE, IWAE, and VampPrior.

**Low-dimensional geometry.** PCA projection of latent samples reveal distinct global organization patterns across models (Figure 5). The VAE latent space forms a single, smoothly connected cloud with strong class overlap. IWAE exhibits sharper cluster boundaries and larger gaps between classes, suggesting more fragmented latent organization. In contrast, VampPrior produces compact and clearly separated clusters already visible, consistent with a structured latent space induced by the mixture-based pseudo-input prior.

**Semantic separability.** Linear probing results confirm that all models encode class-relevant information in a linearly accessible manner. Logistic regression trained on latent representations achieves near-perfect accuracy:

$$\text{VAE: } 0.994, \quad \text{IWAE: } 0.991, \quad \text{VampPrior: } 1.000.$$

This indicates that differences between models are not driven by label separability, but rather by latent geometry and posterior smoothness.

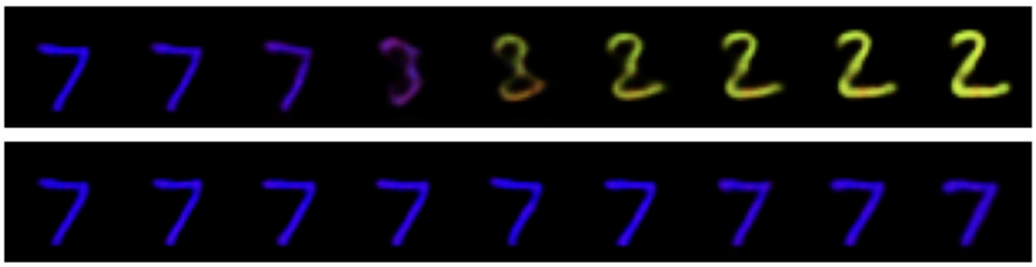
**Local semantic consistency.** Latent traversals further differentiate the models. VAE traversals result in smooth, gradual changes in decoded images, preserving digit identity over wide ranges. IWAE traversals frequently produce abrupt semantic shifts, indicating sharper decision boundaries in latent space. VampPrior traversals are piecewise smooth: transitions are stable within clusters but can change abruptly when crossing mixture components.

**Posterior geometry via MCMC relaxation.** To directly probe posterior smoothness and connectivity beyond variational approximations, we apply MALA relaxation along linear interpolations between latent codes. Table 2 reports segment-wise acceptance rates, serving as a quantitative MCMC diagnostic of local posterior geometry, while Figure 6 provides a qualitative visualization via decoded samples.

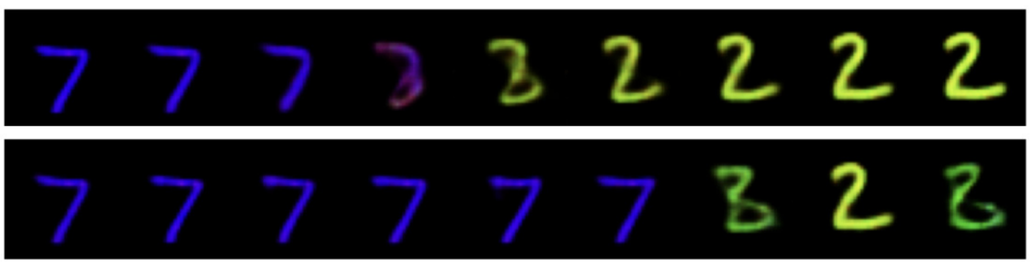
The standard VAE exhibits consistently high acceptance across most segments (typically above $0.85$), with only moderate degradation near interpolation endpoints, indicating a globally smooth and well-connected posterior. In contrast, IWAE shows a pronounced collapse in acceptance rates around the midpoint of interpolation, with values approaching zero, revealing sharply peaked and

Table 2: Segment-wise MALA acceptance rates along latent-space interpolations for different VAE variants. High acceptance indicates locally smooth and well-connected posterior geometry, while near-zero acceptance reveals sharp curvature or disconnected regions.

| Segment | VAE | IWAE | VampPrior |
|---|---|---|---|
| 1 | 0.98 | 0.98 | 0.96 |
| 2 | 0.96 | 0.94 | 0.95 |
| 3 | 0.92 | 0.93 | 1.00 |
| 4 | 0.97 | 0.91 | 0.99 |
| 5 | 0.87 | 0.94 | 0.99 |
| 6 | 0.86 | 0.88 | 0.02 |
| 7 | 0.51 | 0.01 | 0.08 |
| 8 | 0.53 | 0.00 | 0.93 |
| 9 | 0.55 | 0.02 | 0.94 |



(a) VAE



(b) IWAE

(c) VampPrior VAE

Figure 6: Linear latent interpolations (top rows) and corresponding MALA-relaxed trajectories (bottom rows) for all models. Each column corresponds to an interpolation segment between two latent codes.

poorly connected posterior regions likely induced by importance-weighted optimization. The VampPrior VAE demonstrates intermediate behavior: most segments maintain high acceptance, while isolated sharp drops correspond to transitions between mixture components of the learned prior.

Overall, these results highlight how architectural choices directly shape posterior geometry and determine the effectiveness of local MCMC-based exploration.

**Summary.** While all models learn informative latent representations, their posterior geometries differ substantially. VAE provides the smoothest and most stable latent space, IWAE induces overly sharp and locally disconnected posteriors, and VampPrior balances strong global structure with localized discontinuities due to its mixture prior. These findings explain the qualitative differences observed in traversals and MCMC behavior, and highlight posterior geometry as a key factor beyond likelihood-based evaluation.

## 5 Conclusions

All three VAE architectures provide reconstructions of good quality, with the application of Langevin dynamics further enhancing generation performance. Our experiments show that the optimal number of Langevin dynamics' steps should be below $50$ to improve results, otherwise the quality starts to degrade due to excessive perturbations. Beyond reconstruction metrics, our study highlights fundamental differences in the latent space geometry induced by different variational objectives and priors. The standard VAE favors smooth and globally connected representations, IWAE tends to form overly concentrated and fragmented posteriors, and VampPrior induces a structured multimodal latent space aligned with its mixture-based prior. These findings suggest that combining expressive priors with controlled posterior sampling offers a promising direction for improving both generative quality and latent-space interpretability.

## References

Yuri Burda, Roger Grosse, and Ruslan Salakhutdinov. Importance weighted autoencoders. *ICLR*, 2016.

Chris Cremer, Quaid Morris, and David Duvenaud. Reinterpreting importance-weighted autoencoders. *arXiv preprint arXiv:1704.02916*, 2017.

Diederik P Kingma and Max Welling. Auto-encoding variational bayes. *ICLR*, 2014.

Diederik P Kingma, Max Welling, et al. An introduction to variational autoencoders. *Foundations and Trends® in Machine Learning*, 12(4):307–392, 2019.

Gareth O. Roberts and Richard L. Tweedie. Exponential convergence of langevin distributions and their discrete approximations. *Bernoulli*, 2:341–363, 1996. ISSN 13507265. URL `http://www.jstor.org/stable/3318418`.

Jakub Tomczak and Max Welling. Vae with a vampprior. In *AISTATS*, pp. 1214–1223. PMLR, 2018.

Max Welling and Yee W Teh. Bayesian learning via stochastic gradient langevin dynamics. In *Proceedings of the 28th international conference on machine learning (ICML-11)*, pp. 681–688, 2011.